

Open-Vocabulary Object Detection via Vision-Language Knowledge Distillation

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02

Problem & Background

3

Motivation and Objectives

- Closed-vocabulary detectors miss unseen categories.
- Real-world systems require open-vocabulary recognition.
- Leverage vision-language models for rich semantics.
 - Distill V+L knowledge into detectors without labels.
 - Goal: localize and recognize novel categories.
- Bridge closed-world bias; enable open-vocabulary detection.

4

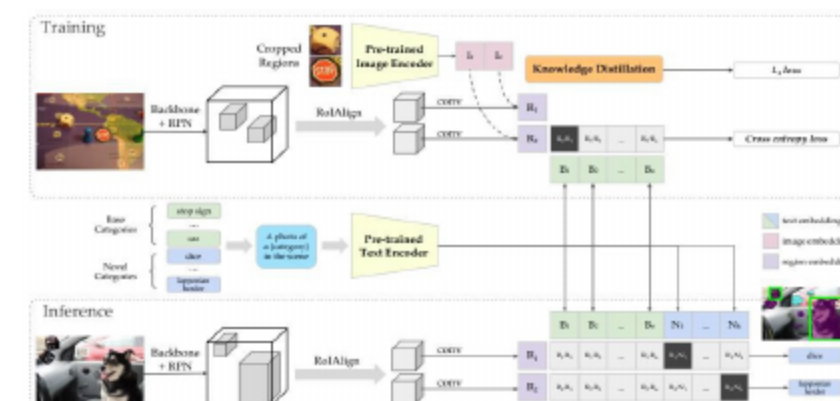
Related Work: Expanding Label Spaces

- Large-vocabulary recognition is a long-standing goal.
- Zero-shot recognition uses attributes, hierarchies, embeddings.
- Early works by Frome (2013) and Norouzi (2014) linked vision to text.
 - CLIP and ALIGN scale image-text learning with strong results.
 - Large-vocabulary detection is costly; ZSD covers few classes.
- ViLD distills image-text teachers, tested on 1,000+ categories.

Method

ViLD: Method Overview

6 ViLD Framework Overview



- End-to-end pipeline: proposals, crops, embeddings
 - CLIP-style supervision aligns vision and text
 - ViLD-text: text embeddings serve classifiers
- ViLD-image: distills a teacher image encoder
 - Cropped-region classification via similarity
 - Unified training boosts novel-category performance

4 ViLD: Variants and Distillation

- Vanilla: learned cls weights; no VL priors, no projection.
 - ViLD-text: L2-normalized text embeddings serve as classifiers.
 - ViLD-image: distill region embeddings from a teacher image encoder.
- Ensembling ViLD-text + open-vocab classifiers boosts robustness.

8 Localization with Cropped Regions

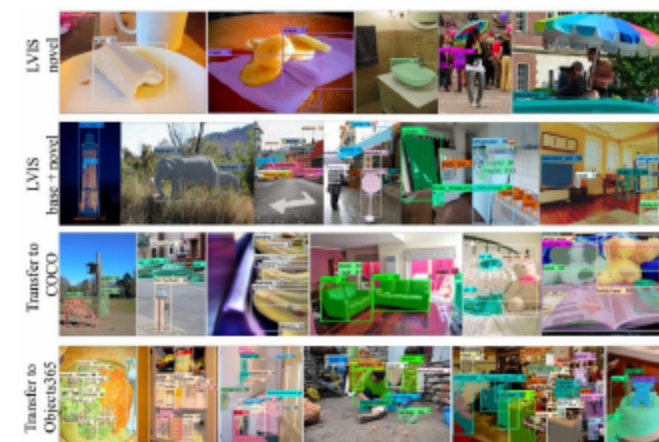


- Modify two-stage detectors to be class-agnostic
 - Generate proposal-driven region crops for RoIs
 - Classify each crop using an open-vocabulary model
- Score categories via text-image embedding similarity
 - Use prompt templates to build text embeddings
 - Improves localization of unseen novel categories

Sec 3

Experiments & Results

10 Qualitative Results and Transfer



- Generalizable region proposals hold across scenes
 - Accurate novel-class labels with minor drop
 - LVIS novel categories detected robustly
- Transfers to COCO and PASCAL VOC with ease
 - Minimal performance loss without tuning
 - Multi-scene results: LVIS novel, COCO transfer

↑ ViLD: Distillation Gains on COCO

- On COCO, ViLD tops zero-shot baselines, no detector pretraining.
 - Distillation boosts confidence and expands category coverage.
 - Text+visual embeddings beat GloVe on novel-category detection.
- CLIP vs Mask R-CNN: side-by-side shows gains on hard objects.

12 Conclusions, Ethics, and Reproducibility

- ViLD distills vision-language knowledge for open-vocab detection.
- Enables generalization to unseen categories in open-world settings.
- Ethics: mitigate dataset bias, fairness gaps, and harmful outputs.
 - Respect privacy; avoid misuse in surveillance or sensitive domains.
 - Reproducibility: document training configs, data splits, and seeds.
- Release code, checkpoints, and evaluation protocols for clarity.